

Track the Eyes

Track the Mind

AC Studio 1 Simulated Realities

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Question



*How can we **get to a destination**
in an unfaimilar environment
if our mobile phone runs out of battery?*

HERE IS WHAT WE DO



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Introduction

Research background

Complexity of urban transportation networks increase cognitive load



<https://www.pexels.com/>

Manley, E. and Cheng, T., 2018. Exploring the role of spatial cognition in predicting urban traffic flow through agent-based modelling. *Transportation Research Part A: Policy and Practice*, 116, pp.554-564.

Wu, J., Chen, X., Zhao, M. and Xue, C., 2024. Cognitive characteristics in wayfinding tasks in commercial and residential districts during daytime and nighttime: A comprehensive neuroergonomic study. *Advanced Engineering Informatics*, 59, 102440.

Research background

Unclear signage prevents pedestrians from obtaining effective path information



<https://www.pexels.com/>

Research background

Lack of contextualized research for
specific objectives



<https://www.pexels.com/>

Research background

Popularity of mobile maps leads to cognitive decline in wayfinding



<https://www.istockphoto.com/>

Mao, Y., Hu, R., Wang, X. and He, W., 2023. Wayfinding efficiency in multidimensional map navigation: A study on spatial cognitive styles and cognitive load. SSRN Electronic
Alinaghi, N. and Giannopoulos, I., 2024. Wayfinding stages: The role of familiarity, gaze events, and visual attention. In: 16th International Conference on Spatial Information Theory (COSIT 2024). Dagstuhl: Schloss Dagstuhl–Leibniz-Zentrum für Informatik.

Research question

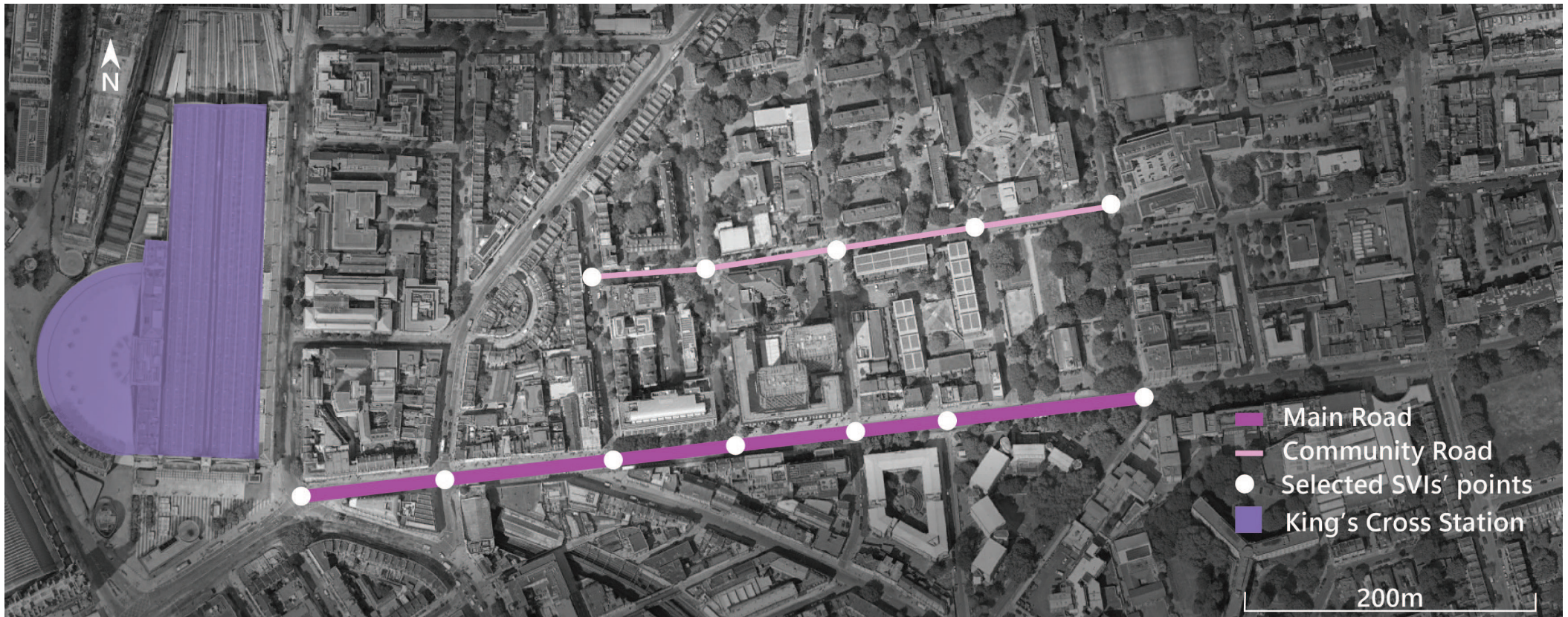
What is the relationship between
pedestrians' visual attention distribution
and *the environmental cues*
they rely on for *wayfinding decisions*
when searching for *a train station*
in static SVIs?

Data Collection

Preparations: Select the Experimental SVIs

Select *4 random images* used before formal test

Use *13 static SVIs* nearby the train station in *formal experiment*

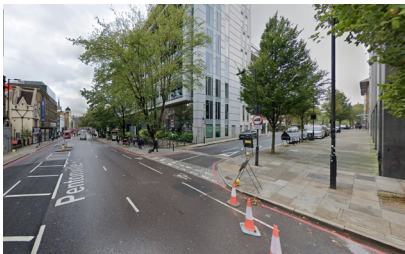


Preparations: Select the Experimental SVIs

Test Images: Random selected



Formal Test: On the main road



Formal Test: On the community road



Preparations: Select Experimental Subjects

Subject One



Aged 23

University Student
Nearsightedness corrected
Relevant background

Subject Two



Aged 22

University Student
Nearsightedness corrected
Relevant background

Subject Three



Aged 23

University Student
Nearsightedness corrected
Irrelevant background

Subject Four



Aged 22

University Student
Ordinary eyesight
Irrelevant background

Subject Five



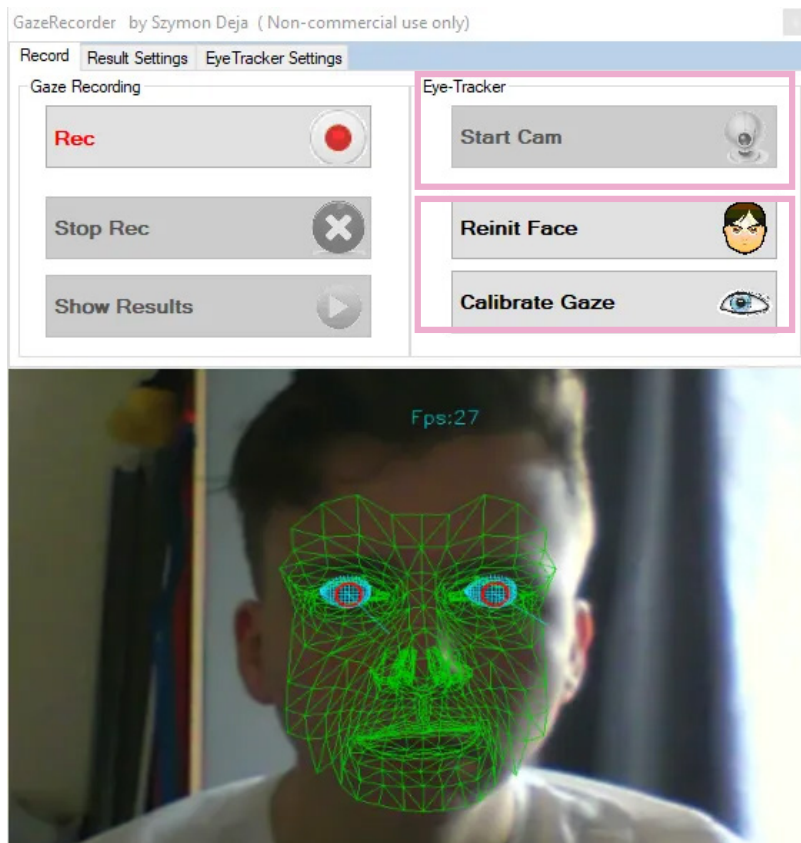
Aged 23

University Student
Ordinary eyesight
Relevant background

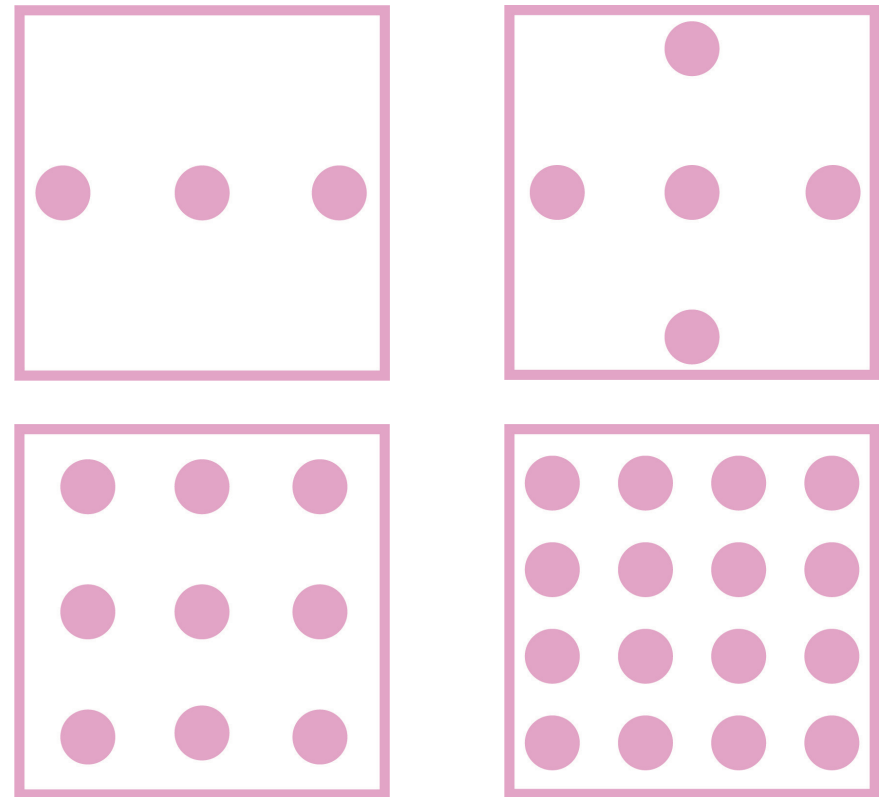
During the Eye-Tracking Experiment

Calibrate the Eye-tracking Tools

Gaze Recorder



Calibration screen with 3, 5, 9, 16 target points



Get the Experiment Results

Use Gaze Recorder to track participants' eyes: *Generate Heatmaps*

Record participants' way-finding choices and reasons: *Voice to Text*



Go Straight

Traffic flow is mainly
on the front



Turn Left

The front feels barren



Go Straight

Wider roads has more pos-
sibility to find a train station



Go Straight

More traffic on the front
way

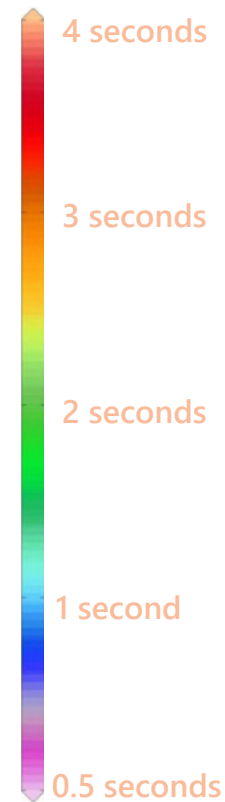
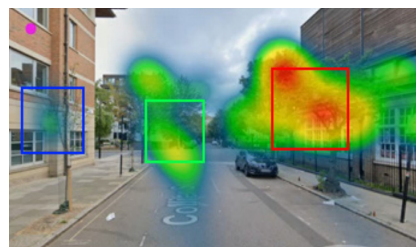
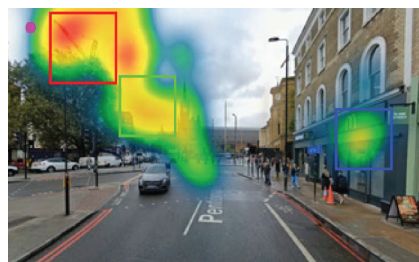
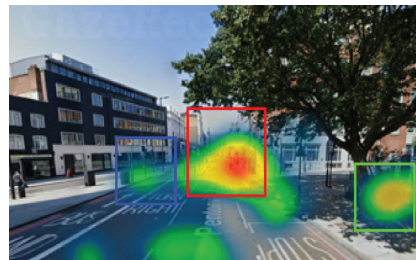
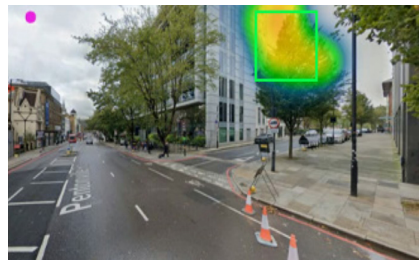
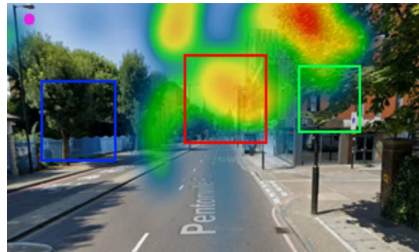
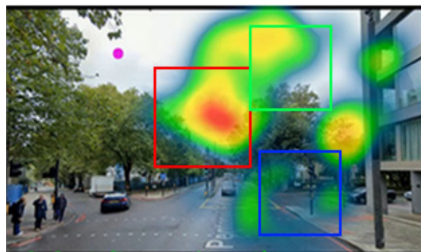


Go Straight

Main road always leads to
a big transportation hub

Data Analysis

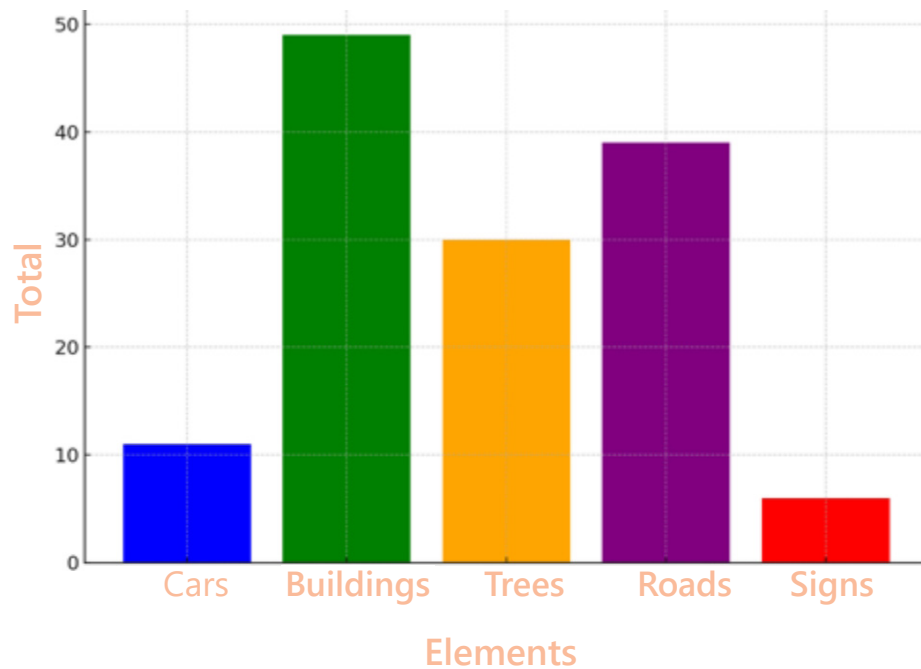
One Subject's Experiment



- Low Focus
- Intermediate Focus
- High Focus

Navigation Experiences Comparison

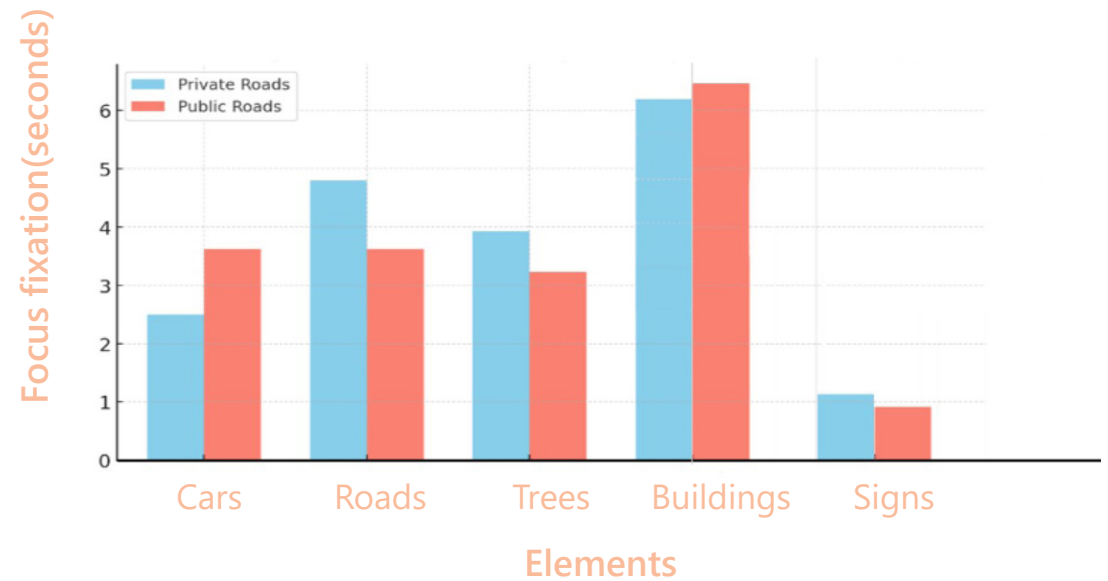
Total distribution of elements



Represents the total visual attention given to different environmental elements across all 13 images in the experiments.

Visual fixation collected in private & public roads images

Points of height on screen



This chart helps to compare the focus fixation between the two types of roads and gives insights into which elements are more significant in each context for way-finding.

Navigation Experiences Comparison

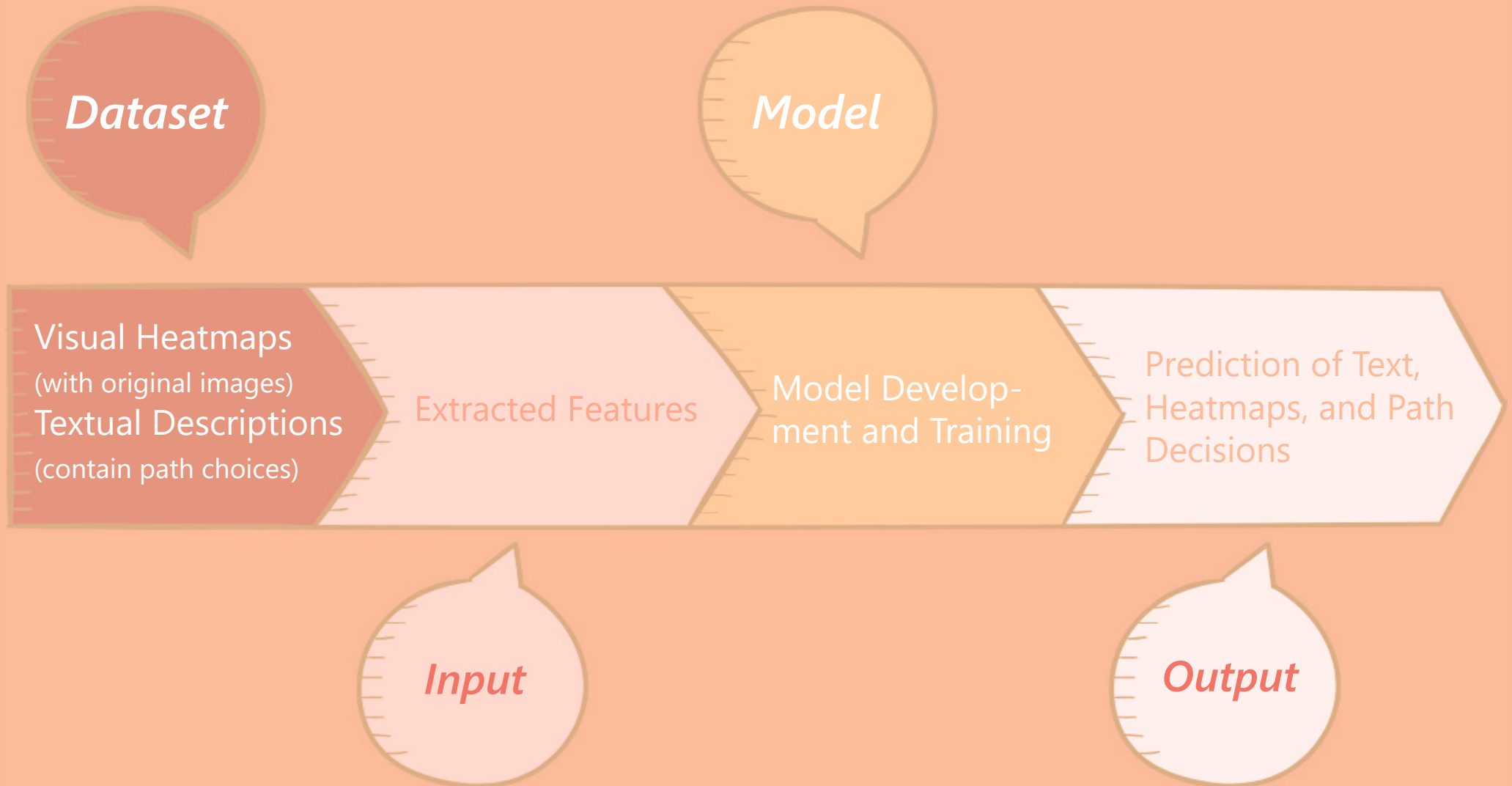
Attention Allocation to Environmental Cues

Elements	1	2	3	4	5	6	7	8	9	10	11	12	13
Cars													
Signs													
Trees													
Buildings													
Roads													

This is table of visual attention data from our eye-tracking experiment, where it shows *what participants' focused the most and least on* and *showing the overall contrast* of the data across various images.



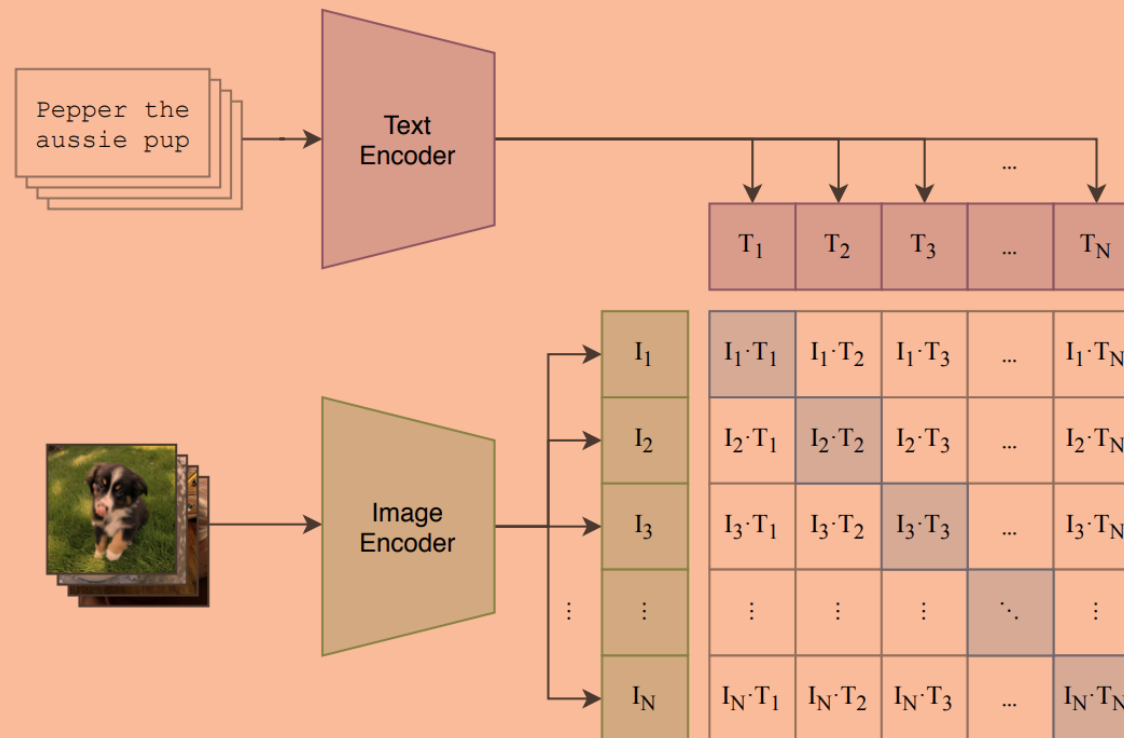
Process Overview: From Raw Data to Prediction



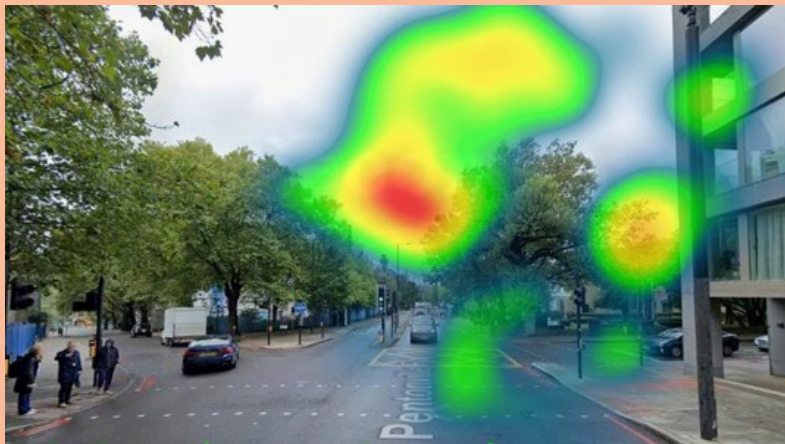
Multimodal Feature Extraction with CLIP

CLIP is being used for its *multimodal capabilities*, mapping visual heatmaps and path descriptions into a shared embedding space.

Contrastive pre-training

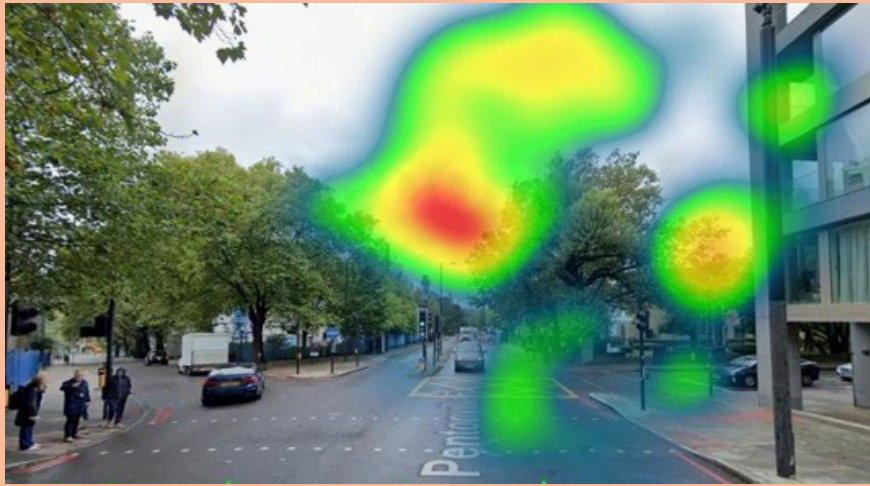


Identifying Hotspot Regions through Visual Comparison

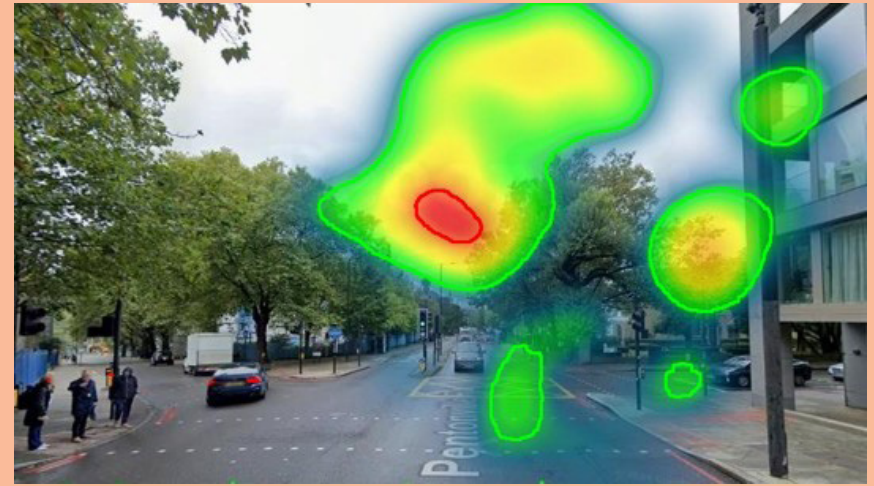


*Compare the original images with the heatmaps, enabling the model to identify which areas are the **focal regions of visual attention***

Leveraging HSV Space for Visual Hotspot Interpretation



Original Heatmap



Colour-Based Simulation

By leveraging the *HSV color space*, the model can better *interpret and simulate the visual hotspot regions* in the heatmap, distinguishing different *hotspot characteristics* more effectively.

Normalization and Punctuation Removal

Tend to go straight ahead. Not many people on the left side, main foot traffic is on the main road.
Straight ahead, no asphalt road on the right, not a carriageway, a footpath.
Straight ahead, no forks in the road in sight, few people on the side roads, lots of people on the main road.
See big sign for building king's cross, will allocate attention to the right side of the road, but still tend to go straight ahead.
Tend to go straight down the main road, more numbers on the right, less traffic, more desolate.
Straight ahead, residential area on the right, landmark building at the end of the road
Straight ahead, the building in front looks like a public station.
Straight ahead, landmark building appears to be a station.
Go left, there are many trees in front of you, no public building seems to be there.
Go ahead, the left side looks like entering a residential area, and the picture of a car traveling straight ahead also influences my decision-making.
Turn left, residential area ahead.
Turn left, the road is wider, trees ahead, looks like I'm going into a residential area
Turn right, both roads feel similar.



few people on left main flow on main road
pedestrian path no asphalt on right not vehicular
few people on small path more on main road
large sign for Kings Cross attention to right road
main road more trees on right few people desolate
residential area on right landmark building ahead
building ahead public station
landmark building seems station
dense trees ahead no public buildings
left side residential area cars going straight
area feels residential
wide road trees ahead likely residential
both paths similar

Text normalization and punctuation removal ensure **consistency** in textual data by **standardizing expressions** and simplifying structure.

This preprocessing step creates **cleaner features**, improving model performance and accuracy.

From Raw Data to Meaningful Features

Feature Extraction

Original Input Features



Create new features
through linear
combination



Final created features

Feature Extraction converts raw data (e.g., text or images) into *fixed-dimensional vectors* with meaningful, task-relevant information.

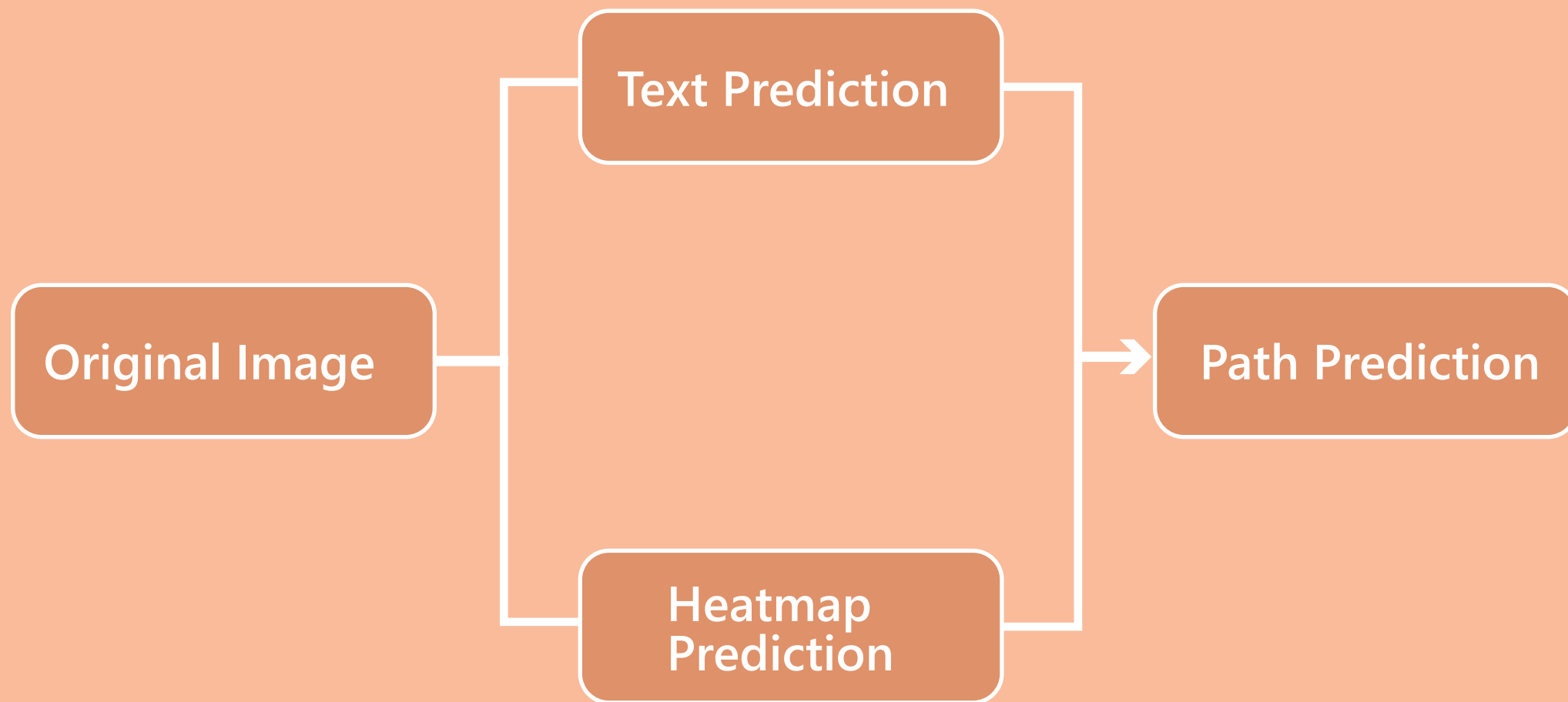
Bridging Heatmaps and Path Descriptions



Compared the similarity between the two sets of features using
cosine similarity

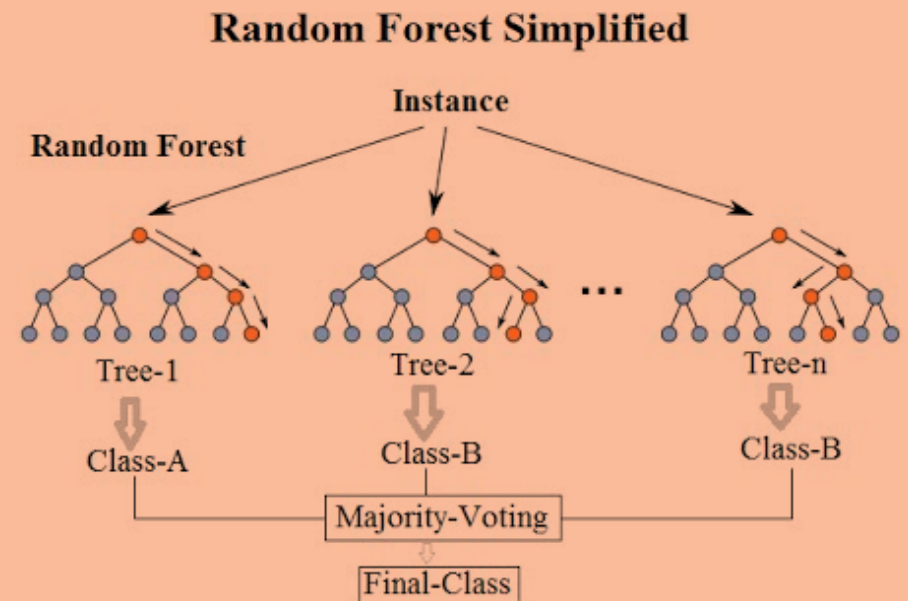
Helping the model analyze their *semantic relationship* as a step toward predicting navigation behavior.

Model Workflow Overview



Path Decision Modeling: Combining CLIP and Random Forest

Using the *CLIP model* and *Random Forest* classifier to analyze the impact of visual hot-spots and textual descriptions on path choices.



Path Decision Modeling: Achieving 80% Accuracy

The model achieved **80% accuracy**, improving from the 50% baseline.

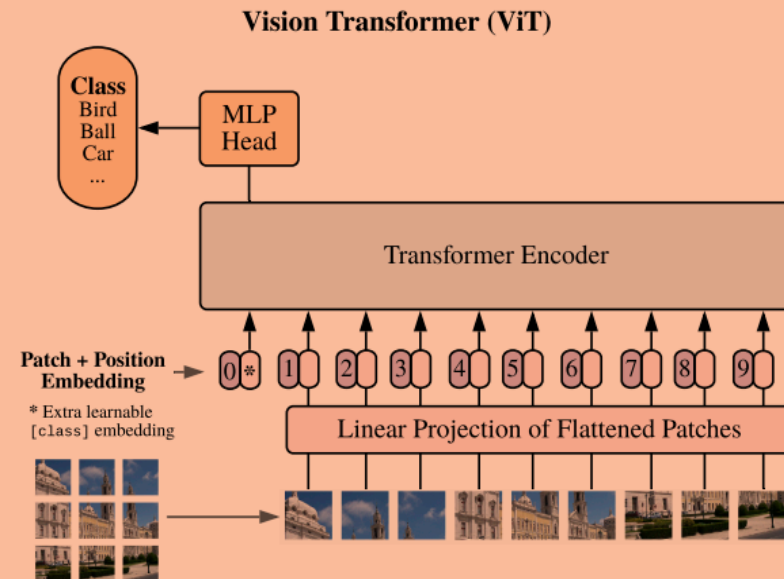
This demonstrates the value of combining textual and visual features with the Random Forest Classifier, showing potential for further optimization.

```
Resampled data shape: (69, 1024) Number of labels: 69
Cross-validation accuracy scores: [0.82608696 0.91304348 0.86956522]
Mean cross-validation accuracy: 0.8695652173913043
Random Forest model accuracy (using test set): 0.7857142857142857
Random Forest confusion matrix:
[[2 1 2]
 [0 7 0]
 [0 0 2]]

Process finished with exit code 0
```

Heatmap Prediction Modeling: Vision Transformer

```
26 heatmap_weights = [2.0, 1.5, 1.0]
27
28 # Load dataset
29 dataset = ExtendedImageDataset(original_folder, heatmap_folders, transform=data_transform, weights=heatmap_weights)
30 train_loader = DataLoader(dataset, batch_size=4, shuffle=True)
31
32 # Model setup
33 vit_model = ViT(
34     image_size=224,
35     patch_size=16,
36     num_classes=224 * 224 * 3, # Output is the shape of the original image
37     dim=1024,
38     depth=6,
39     heads=16,
40     mlp_dim=2048,
41     dropout=0.1,
42     emb_dropout=0.1
43 ).to(device)
44
45 # Add a decoder to the ViT model to restore the output to image shape
46 class ViTWithDecoder(nn.Module):
47     def __init__(self, vit):
48         super(ViTWithDecoder, self).__init__()
49         self.vit = vit
50         self.decoder = nn.Sequential(
51             nn.Linear(224 * 224 * 3, 224 * 224 * 3),
52             nn.ReLU(),
53             nn.Linear(224 * 224 * 3, 224 * 224 * 3),
54             nn.Sigmoid()
55         )
56
57     def forward(self, x):
58         x = self.vit(x)
59         x = self.decoder(x)
60         x = x.view(-1, 3, 224, 224) # Restore to heatmap shape
61         return x
62
63 model = ViTWithDecoder(vit_model).to(device)
64
65 # Loss function and optimizer setup
66 criterion = nn.MSELoss()
67 optimizer = optim.Adam(model.parameters(), lr=0.001)
68
69 # Train the model
70 for epoch in range(20):
71     model.train()
```



ViT Transformer is selected for handling color information in images, effectively processing high-dimensional data and using *attention mechanisms* to capture color differences in heatmap areas.

Limitations & Improvements

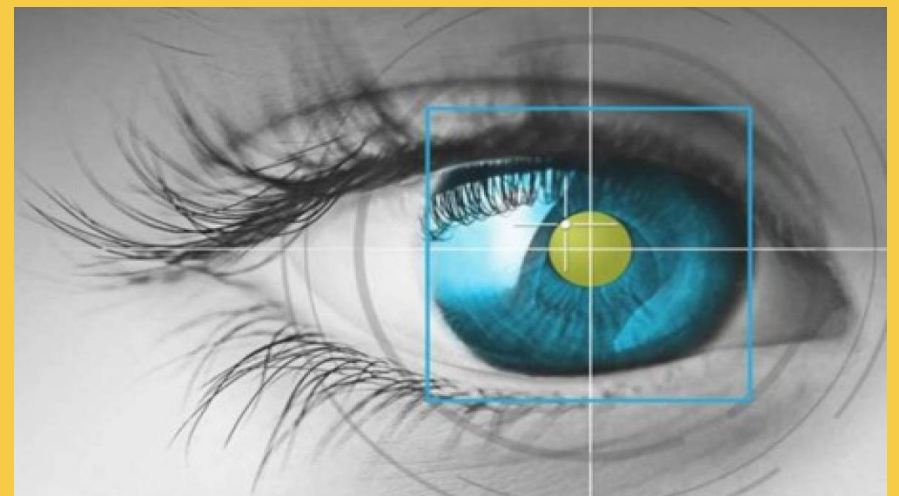
Limitations

Inaccurate eye-tracking experiments



Improvements

Run the customized openCV code



Limitations

Low screen resolution influences the recognizability of SVIs

Improvements

Try to use another software whose resolution can be adjusted



800*600



1427*887

Limitations

Restricted information of static SVIs →



Static Test Image

Improvements

Run the experiment in interactive 3D navigation



Surroundings

Limitations

Homogenous background of experimental subjects



Improvements

Expand the population categories instead of limited the Uni Students



References

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THANKS

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